System Modeling, Coupling Analysis, and Experimental Validation of a Proportional Pressure Valve With Pulsewidth Modulation Control

Fei Meng, Hui Zhang, Dongpu Cao, and Huiyan Chen

***Abstract*—This paper presents system modeling and coupling analysis of a proportional solenoid valve (PSV) with pulsewidth modulation (PWM) control mode. The coupling analysis is adopted to reveal the internal relations among the electromagnetic, mechanical, fluid, and electrical subsystems based on the integrate proportional pressure valve mode for the PSV system. Then, a simplified and practical PWM strategy is proposed to design the PSV. The dynamic performance of each subsystem is analyzed with respect to a duty cycle set. Consequently, simulation using a commercial software is presented. It infers from the results that the valve has a fast response to the control signal and can reach a small flow area at the instantaneous moment during opening and closing operations. Finally, the analytical results of the designed PSV are validated by using experimental tests. It follows from the testing results that the established model is precise and the corresponding coupling analysis results are valuable for the performance design and optimization of PSV.**

***Index Terms*—Coupling analysis, experimental validation, proportional solenoid valve, system modeling.**

I. Introduction

HIGH precision and fast response performances are critical for actuators in vehicle systems [1]–[3]. The design and precision control of vehicle actuators have drawn great attention during the past few years; see [4]–[6] and the references therein. The demand for fuel economy and exhaust emissions reduction calls more highly efficient engines and multispeed transmissions [7], [8]. Hydraulic systems have been widely used in vehicles due to the significant properties such as high power density, flexibility, and high stiffness [9]–[11]. Among the hydraulic systems, solenoid valves are employed to enhance transmission shift quality, improvement engine efficiency, and increase fuel economy [12]–[14].

The proportional solenoid valve (PSV) provides a particular output pressure which is directly proportional to the current applied to the solenoid coil in the valve [15]–[17]. To realize the proportional property, the PSV system consists of interacting electrical, electromagnetic, mechanical, and fluid dynamical subsystems. All of these subsystems are strongly coupled together. Due to the complexity for the PSV in the structure, several challenges have been found in present applications. The challenges include unsteady electro-dynamical operation, unsteady fluid mechanical forces on the valve body, and time- and pressure-dependent fluid transmission characteristics [18], [19]. Therefore, it is quite difficult to investigate the detailed and accurate features of the PSV system.

Some simulation methodologies have been developed for proportional pressure valves in the past. Many studies dealt with the multiscale modeling and coupling simulation, which can provide insight into magnetic fields with a combined 3-D and 1-D model approach. There are also many studies in which the modeling and control of proportional valves are considered at the system model level. Generally, researchers have treated the modeling of solenoid valves completely through mathematics or using bond graph simulation technique [20]–[22]. As the magnetic field is strongly coupled with the other subsystems, a precise mathematical model of the magnetic field cannot be easily established. Wang *et al.* in [10] identified the flux linkage characteristics using combined dynamic data from the measurements of the magnetic force at different positions and the currents from steady-state experiments. Due to the complexities of the magnetic field and the flow field, one of the most promising approaches is to develop numerical methods to analyze the magnetic and flow subsystems. Among the numerical methods, the finite-element method (FEM) has been widely adopted in the PSV analysis [23]–[25]. However, the magnetic field and the flow field were analyzed. In order to reduce the cost and understand electromagnetic actuator systems, it is necessary to perform multiphysics simulations in which both the electromagnetic and hydraulic forces are considered. In the study [26], the authors presented a multidomain nonlinear 1-D dynamic model of a PSV. Comparison analysis was adopted to validate the simulation model where can be obtained for the proposed model. Sorli *et al.* [27] developed a three-way pneumatic proportional valve actuated by a proportional

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solenoid with the pulsewidth modulation (PWM) control. Lee *et al.* [28] also gave an insight into a linear controlled solenoid valve with multiphysics simulation methodology, and the experiments were performed to verify the reliability of the simulation results. In the above studies, the effects of the driving circuit were not incorporated. As the solenoid resistance changes over a wide range and the relationship between the PWM duty ratio and the solenoid current varies when the solenoid resistance is changed, the analysis of driving circuit is necessary for the system's entire performance. For the analysis of driving circuit, Najjari *et al.* in [29] proposed a position control method for an electro-pneumatic system based on the PWM technique. The solenoid driving circuits were modified in [30] and [31] to improve the solenoid response performance. In addition, Taghizadeh *et al.* in [32] investigated the static characteristics of a PWM solenoid valve and presented an equation for determining the maximum operating modulation ratio.

Though there are a lot of studies on the coupling analysis of a PSV system in the existing literature, the fluid dynamical subsystems are different from each other. Since a PSV system has four coupled subsystems and each subsystem plays an important role for the entire system, it is desired to analyze the four coupled subsystems simultaneously to reveal the effects to the overall performance. In addition, for heavy-duty automatic transmissions, the clutch actuation system should have a fast response and a high flow capacity simultaneously [33]. Thus, a high-speed PSV pilot operated pressure reducing valve is essential for clutch actuation systems. These motivate us to do the research in this work.

We aim to carry out a comprehensive coupling analysis for a predesigned PSV system by using multiscale modeling and a coupled simulation technique. A nonlinear 1-D system model which reveals the electrical, mechanical, and fluid dynamics physical characteristics of the valve based on the physical laws has been developed. Moreover, a 3-D model of the magnetic field is established by using FEA code and the coupling effects of the other subsystems are discussed. The PSV simulation results are validated with experiments. In addition, some extreme operations in practical aspects are evaluated. The outline of the study is organized as follows: In Section II, the PSV structure and operation principle are introduced. Then, in Section III, we establish a comprehensive model consisting of an electrical subsystem, an electromagnetic subsystem, a fluid dynamical subsystem, and a mechanical subsystem. In Section IV, the comprehensive model is employed to the coupling analysis and the analysis results are more reasonable. In Section V, experimental tests are carried out to validate the previous analysis results.

II. PSV Structure and Operation Principle

The solenoid valve under study in this paper is a pressure reducing/relieving valve which can be infinitely adjusted across a prescribed range by using a variable electric input. Fig. 1 illustrates the cross-sectional view of a PSV used in automatic transmissions. The PSV is a normal closed valve, which has two subsystems named electromagnetic and fluid part. In the electromagnetic section, the dot line shows the flux path while the solid line shows the flow path of the fluid in the fluid section.

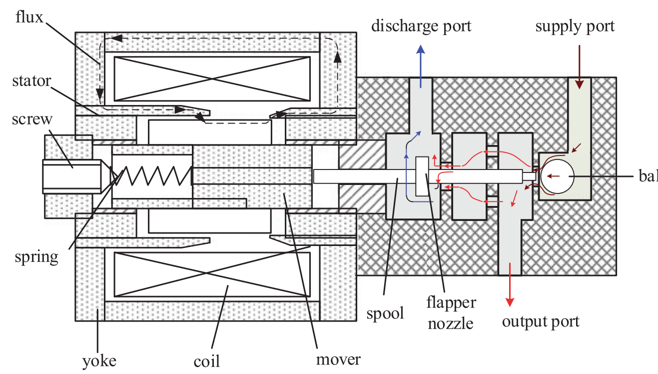


Fig. 1. Cross-sectional view of the PSV structure.

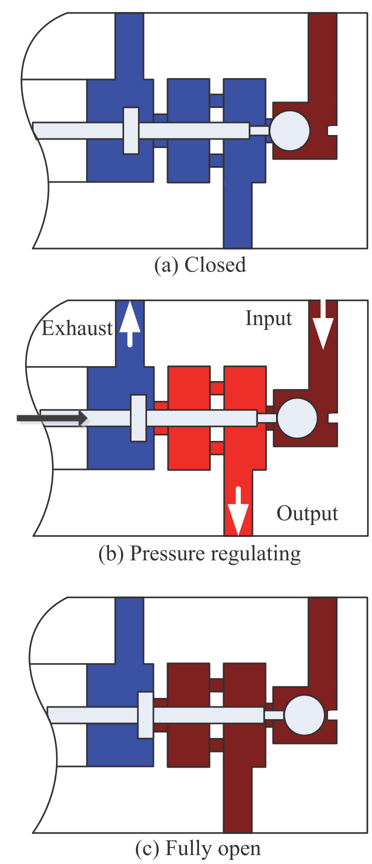


Fig. 2. Operation principle of the proportional solenoid value. (a) Closed. (b) Pressure regulating. (c) Fully open.

Fig. 2 shows the operation principle of the proportional solenoid value. The valve operates as follows: 1) When the coil is not energized, the spring force is balanced by the fluid force making the spool remain at the initial position. The valve is fully closed as shown in Fig. 2(a). 2) When applying a current through the coil, a thrust is generated and the mover is attracted toward the stator in the direction indicated by the arrow as shown in Fig. 2(b). The mover is connected with the spool, the generated thrust will exert on the control valve. The input port is opened and the fluid flows into the chamber of the PSV; the fluid in the chamber not only flows into the output port but also flows toward the exhaust port. The opening degree of the flapper which is connected to the exhaust port can control the output pressure indirectly. 3) When the current is still increasing, the exhaust port will be closed due to the increased electromagnetic force. The output pressure will reach the maximum value which is determined by the input pressure as shown in Fig. 2(c).

The spring force offsets a portion of the fluid pressure so that slight electromagnetic force can open the valve quickly. The spring force also realizes the relief function when the supply pressure fluctuates and limits the output pressure. These types of solenoid valves are referred to as proportional bleed valves since a relatively low fluid flow through the hydraulic portion of the valve were adopted.

III. PSV System Modeling

The system model is important for the coupling analysis. At the present state of investigation, we develop a comprehensive system model and the model is devoted to electrical, electromagnetic, mechanical, and fluid dynamic subsystems. Such system modeling generates a future basis for a precise distinction between the electromagnetic and fluid mechanical effects influencing the valve body motion. Before proceeding, the following assumptions are made:

1) A constant supply source to the valve inlet port is assumed.
2) The spring is assumed to be linear.
3) With the opening and closing of the valve ports, the dynamic flow force acting on the valve spool is neglected and only the steady-state flow force is considered.
4) Stick-slip friction and coulomb friction effects are ignored.

In the automatic transmissions, a pressure regulating valve will realize a constant supply pressure. The linear spring assumption is suitable for the small displacement of the mover. Considering the small dimension of the spool, the dynamic flow force act on the valve and the friction force are very small in comparison to other forces, thus these forces can be ignored.

A. Electromagnetic Subsystem Modeling

Electromagnetic subsystem is the key component of the PSV. The electrical signal is converted into a linear mechanical force signal according to its special structure. The analysis of an electromagnetic field by using an FEM follows the assumptions in [16]. The Maxwell's equation can be expressed as follows:

$$\nabla \times H = J \tag{1}$$

$$\nabla \times E = -\frac{\partial B}{\partial t} \tag{2}$$

$$\nabla \cdot B = 0 \tag{3}$$

$$B = \mu\mu_0 H \tag{4}$$

$$J = \sigma E \tag{5}$$

where B is the magnetic flux density (T), H is the magnetic field (A/m), J is the externally generated current density vector (A/m2), E is the electric field intensity (V/m), σ is the electrical conductivity (S/m), μ is the relative magnetic permeability, and μ_0 is the permeability of free space $\mu_0 = 4\pi \times 10^7$(H/m). By using the constitutive relation and the electric field relation, the Ampere's law of the electromagnetic field can be expressed as

$$\sigma \frac{\partial A}{\partial t} + \nabla \times \left(\frac{1}{\mu\mu_0} \nabla \times A\right) = J \tag{6}$$

where A is magnetic vector potential. When the specifying current in the solenoid coil is denoted by $i(t)$, the out-of-plane component of the current density J_0 is defined as

$$J_0 = \frac{Ni(t)}{A_{\text{coil}}} \tag{7}$$

where N is the specified number of coil turns (turn) and A_{coil} is the total cross-section area of the coil domain (m^2).

The PSV uses three different magnetic materials. The magnetic materials are high-permeability soft magnetic iron (DT4C) in the mover, machine structural carbon steel (ASTM 1020) in the stator, and machine structural carbon steel tubing (ASTM 1010) in the yokes. As the mentioned materials are nonlinear materials, the relationship between the magnetic field H and the magnetic flux density B is also nonlinear.

B. Mechanical Subsystem Modeling

The mechanical subsystem consists of a mass, spring, and damper under the effects of magnetic and pressure forces. This subsystem links the electromagnetic and fluid flow subsystem. The subsystem can be represented using Newton's second law

$$m\ddot{x} = F_{\text{mag}} - k_b\dot{x} + k_s(x + x_0) - F_{\text{fluid}} \tag{8}$$

where x is the position of the spool, x_0 is initial compression of the spring, F_{mag} is the electromagnetic force, F_{fluid} is the fluid pressure force acting on the spool, k_sx is the combined spring force, k_s is the spring constant, $k_b\dot{x}$ is the viscous damping force, and k_b is the viscous coefficient which can be calculated as

$$k_b = \frac{2\pi \cdot u \cdot d_p \cdot l_c}{d_c} \tag{9}$$

where d_p is the external piston diameter, u is the absolute viscosity of the fluid, d_c is the clearance on diameter, and l_c is the length of contact.

C. Fluid Dynamical Subsystem Modeling

The fluid dynamical subsystem can convent the mechanical force signal into the fluid signal (output) finally. We can see from Fig. 3 that the supply port may be considered to be a ball poppet valve as it has the characteristics of a switch. The discharge port can be considered to be a single flapper nozzle valve since the distance between the nozzle and the flapper will change with the spool movement. This will ensures the control of the outflow rate.

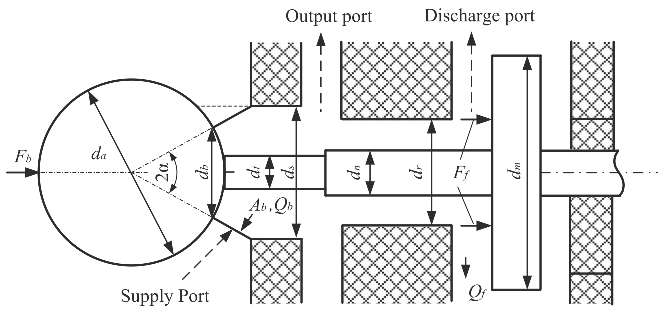


Fig. 3. Fluid subsystem.

The flow rate of the ball valve is then

$$Q_b = C_b \cdot A_b \cdot \sqrt{\frac{2\Delta P}{\rho}} \tag{10}$$

where C_b is the flow coefficient, A_b is the flow area, and $\Delta P = P_s - P_k$ is the differential pressure. P_s is the supply pressure, and P_k is the output pressure. In this paper, all pressures are gauge pressures.

The pressure force acting on the ball can be expressed as follows:

$$F_b = P_s \cdot \frac{\pi}{4}({d_a}^2 - {d_t}^2) - P_k \cdot \frac{\pi}{4}({d_s}^2 - {d_t}^2). \tag{11}$$

The flow rate of the 1-D motion of a nozzle poppet valve can be expressed as

$$Q_f = C_f A_f x \sqrt{\frac{2 \cdot P_k}{\rho}} \tag{12}$$

where A_f is the throat area of the annular orifice, which can be expressed as $\pi(d_m^2 - d_n^2)/4$. Here, the discharge port pressure is zero. C_f is the valve discharge coefficient.

The fluid force acting on the flapper F_f can be expressed as [34]:

$$F_f = P_k A_f \left[1 + \frac{16C_f^2}{(d_r - d_n)^2}(x_{\max} - x)\right] \tag{13}$$

where $x_{\max}$ is the maximum displacement of the spool valve.

By neglecting the leakage, the output flow rate of the PSV can be expressed as

$$Q_o = Q_b - Q_f \tag{14}$$

where Q_o is the output flow rate.

D. Electrical Subsystem Modeling

For engineering applications, the PSV is controlled by the PWM signal from the transmission control unit. The control signal for driving the PSV is amplified by an electrical subsystem. The PSV coil is ideally represented as an inductor in series with a resistor. In this case, the voltage drop across the circuit is expressed using the flux linkage and the coil resistance *R*. The equation for the electrical circuit is then

$$U = Ri + \frac{d\psi}{dt} \tag{15}$$

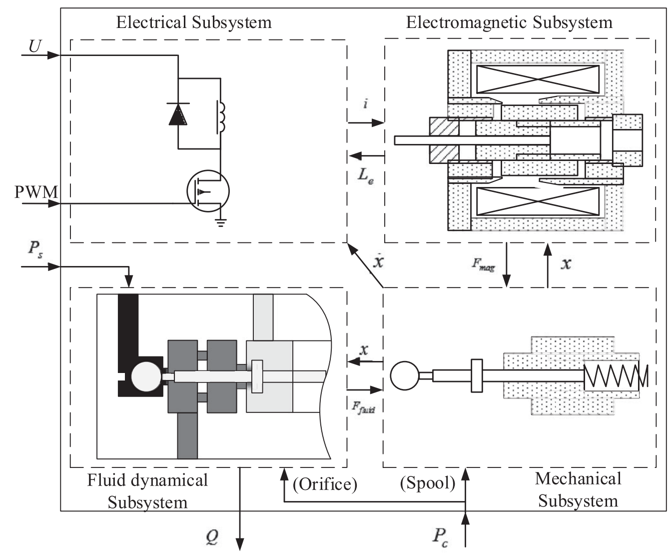


Fig. 4. Interactions among the subsystems for the PSV.

where *i* is the current. The flux linkage is not fully coupled to the magnetic circuit and is dependent on the current of the coil and the air-gap distance x_{air} which can be influenced by the valve displacement *x*. (15) can be expanded as [32]:

$$U = Ri + \left(L_e + \frac{\partial\psi}{\partial i}\right) \cdot \frac{di}{dt} + \frac{\partial\psi(x_{\text{air}}, i)}{\partial x_{\text{air}}} \cdot \frac{dx_{\text{air}}}{dt} \tag{16}$$

where an external inductance term L_e represents the leakage.

A PWM controlled solenoid can reduce the price of the actuator and the total system. The initial current of the nth PWM ON and OFF state can be represented as [35]

$$I_{0+}(n) = \frac{U}{R}\left(e^{-\frac{(1-\delta)T}{\tau}} - e^{-\frac{T}{\tau}}\right)\frac{1 - e^{-\frac{(n-1)T}{\tau}}}{1 - e^{-\frac{T}{\tau}}} \tag{17}$$

$$I_{0-}(n) = \frac{U}{R}\left(1 - e^{-\frac{\delta T}{\tau}}\right)\frac{1 - e^{-\frac{nT}{\tau}}}{1 - e^{-\frac{T}{\tau}}} \tag{18}$$

where $I_{0+(n)}$ denotes the initial current of nth PWM ON state and $I_{0-(n)}$ denotes the initial current of the nth PWM OFF state. δ is PWM duty cycle and has a value ranging from 0 to 1. *T* is the period of PWM driving signal. τ is the electrical constant.

IV. PSV System Coupling Analysis

A. Interrelationship of the Subsystems

A block diagram representation of the valve is given in Fig. 4, which indicates the signal flow of the model from the input voltage (*U*), supply pressure (P_s), and PWM control signal to the output flow. The arrows in the box denote the intermediate interactions among different subsystems. In the solid box, there are four dashed boxes which represent the electrical, electromagnetic, mechanical, and fluid dynamical subsystems. The interactions between each subsystem can be observed in Fig. 4 and are explained one by one in sequel.

First, in the electrical and electromagnetic subsystems, when changing the current (*i*), the magnetic flux in the magnetic field will be affected. The magnetic flux change will directly affect the electromagnetic force ($F_{\rm mag}$). The spool moves will cause the inductance (L_e) change in the winding. This will affect the PWM control input which determines the necessary current.

Second, for the electromagnetic and mechanical subsystems, the electromagnetic force will act on the spool and cause the spool to move. The spool displacement (x) will affect the magnetic field characteristics.

Third, between the mechanical and fluid dynamic subsystems, the fluid pressure force ($F_{\rm fluid}$) will exert on the spool, which will balance with the electromagnetic force and the spring force. The combined forces would move the spool by a distance (x). The fluid force acting on the spool will change because of the spool movement.

Finally, the changing speed ($\dot{x}$) of the spool movement will generate an induced electromotive force in the winding, which will also affect the PWM control input.

In the next section, the nonlinear dynamic equations governing these subsystems will be derived. The pressure forces will also be included. The winding of the PSV in the electric control circuit can be simplified as a resistor and an inductor. The electromagnetic subsystems, namely the solenoid with its electromagnetic equations, and the mechanical part consisting of a mass-spring-damper system, behave dynamically and have transient responses.

B. Analysis Method and Procedure

A feedback loop iteration procedure is adopted to solve the coupling analysis problem. A schematic diagram of the calculation procedure is shown in Fig. 5.

First, the steady state ($t = 0$) is determined: the pressure and the flow for steady flow condition are calculated. The dynamic forces, except for the spring force and the pressure force, act on the ball poppet, the winding current, and valve displacement. All other variables are zero in this case. Only the spring force and the pressure force of the ball poppet exert on the spool. Then, introducing a control command ($t > 0$), the overall spool behavior in the linear control solenoid valve is determined step by step as follows:

1) The time-varying PWM duty cycle is given by the controller.
2) The control signal is amplified by the driving circuit and the current applied to the PSV magnetic field can be calculated using (16) and (17), which will change the intensity of the magnetic field H.
3) Solving the electromagnetic subsystem using (1) and (7) and the fluid dynamic subsystem using (11) and (13) simultaneously but separately. The electromagnetic force and the mechanical force, including pressure force, spring force, and viscous damping force are applied to the spool. The communication frequency between the electromagnetic subsystem and the mechanical subsystem is consistent with the fluid dynamic and mechanical subsystems.
4) The response analysis of this mass spring damper system decides the next position of the spool, while the spool displacement, velocity, and acceleration feed back to the electromagnetic and fluid dynamic system.
5) Proceeding to the next time step $t = t + \Delta t$ and repeat the process from 1).

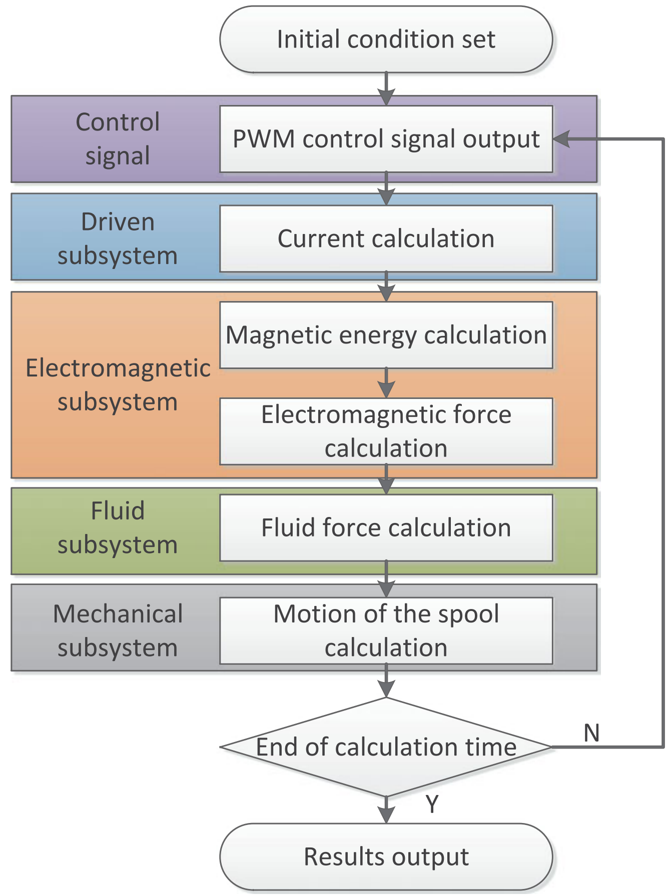
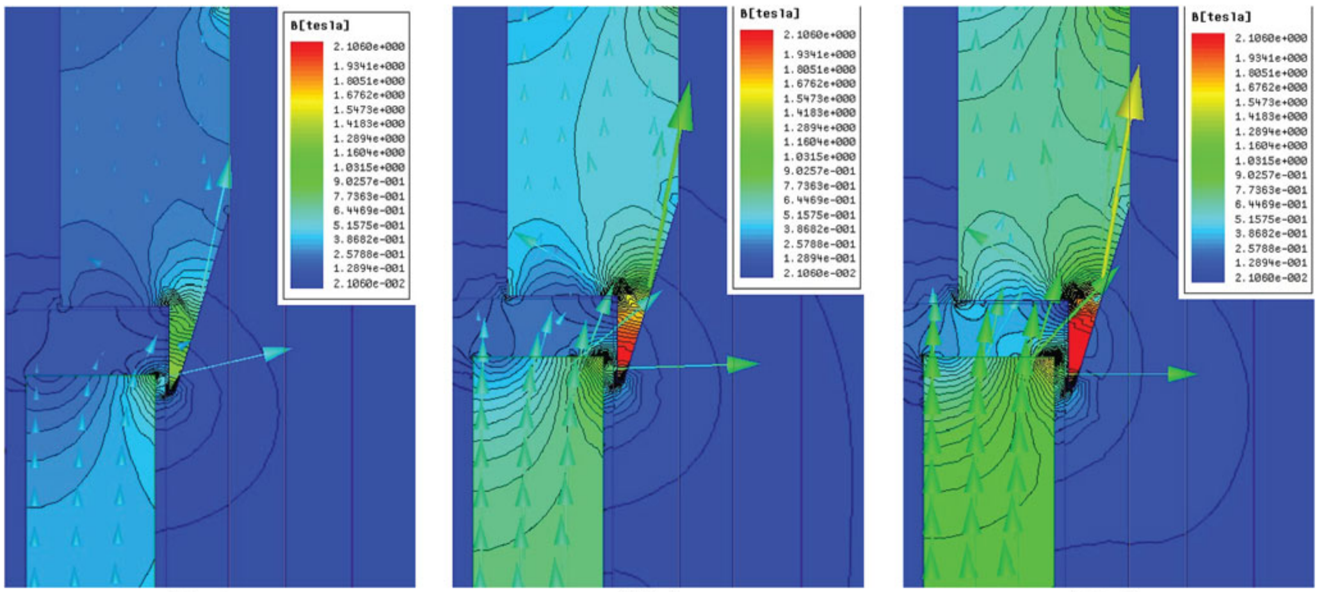


Fig. 5. Schematic diagram of the calculation procedure.

C. Illustrative Numerical Example

In order to clearly understand the PSV system performance and the dynamic variations of the internal parameters when the PSV is opened and closed controlled by the PWM signal, a numerical example is used to illustrate the procedure. Different from the work in [36], the numerical example of this paper not only concerns the opening stage of the PSV, but also the closing stage. The PSV PWM-driving condition is adopted from the high-side and low-side method. For the low-side driver, the transistor (switch) is between the solenoid and ground. Compared with high-side driver, the low-side driver is easy to implement but may be less safe. This paper will adopt the low-side to control the PSV system, as shown in Fig. 4. The PWM frequency is set to 1000 Hz. The entire simulation process takes 80 ms and is divided into four stages: first, the duty cycle varies from 0% to 100%, which lasts 32 ms and has a growth rate of 1%; on the second stage, a 100% duty cycle is maintained for 8 ms; on the third stage, the duty cycle ranges from 100% to 0% of



(a) t=2ms (b) t=3ms (c) t=15ms

Fig. 6. Electromagnetic flux densities and vectors in the transient electromagnetic simulation. (a) $t = 2$ ms. (b) $t = 3$ ms. (c) $t = 15$ ms.

the decreased rate of 1%, and the last stage includes holding a 0% duty cycle for 8 ms. The involved parameters and boundary conditions used in the coupled simulation of the PSV are shown in Appendix A (Table I).

The coupling methodologies can be divided into manual ones or automatic ones depending on the method of data transfer between the two codes. In this study, the coupling model involved a 3-D model (for the electromagnetic system), three 1-D models (for the mechanical, fluid flow, and electrical system), and control signal within the overall model. Because the proportional solenoid is axisymmetric, a 2-D axisymmetric analysis is performed with the ANSYS Maxwell. The mover with its surrounding air are meshed. For any plunger position, the model is meshed separately and boundary conditions are applied. Thus, the magnetic force is obtained for any position in a separate meshed model. And other 1-D models are solved by MATLAB/Simulink.

In the magnetic field, axially symmetrical governing equations are employed and the far field boundary condition is the magnetic insulation. In the flow field, the discharge gauge pressure is atmospheric pressure. The inlet boundary condition of the feedback pressure is assumed to be equal to the output pressure and is varied at every time step. The normal operation temperature for all systems is chosen as 20 °C.

D. Numerical Example Simulation Results

1) Electromagnetic Field: Fig. 6(a) to (c) depicts the local magnetic field diffusion process for $t = 2$ ms, $t = 3$ ms, and $t = 15$ ms, respectively. The magnetic flux flows into the valve through the stator and is again circulated through the yoke. The electromagnetic force is determined from the amount of magnetic flux which passing through the mover. At $t = 2$ ms, most of the magnetic flux passing through the main air gap, as shown in Fig. 6(a). The region near the tip of the stator is relatively weak and generates a smaller axial force. With the time goes up, the magnetic flux passing through the valve wall increases and an increment in electromagnetic force occurs. As shown in Fig. 6(b), when $t = 3$ ms, the current changes caused by the region near the tip of the yoke are highly saturated. The permeability of the highly saturated region is close to that of the surrounding air. Thus, the dispersed flux heading toward the less saturated region has an axial component that contributes to axial force generation. This makes the valve move. Fig. 6(c) shows, when $t = 15$ ms, the valve has reached its maximum displacement (see Fig. 8). For the minimum air gap, most of the total flux bypasses the stator due to its permeability. The flux then increases as the current is increasing and most of flux passes through the stator. Therefore, the electromagnetic force still increases (see Fig. 9(a)).

Fig. 7 describes the variation in amplitude of the coil current, the coil inductance, the flux linkage, and the induced voltage during the PWM working cycle. Before the electromagnetic force reaches an amount previously sufficient to move the spool, the coil current is gradually increased, which causes a flux increase in the spool. The spool displacement and the third term in (16) are both zero. The current will be found to grow exponentially when the equation is solved. The average current verified the initial growth stages. The flux linkage change trend is consistent with the current because the flux linkage is proportional with the current. The initial value of the inductance is approximately 37 mH, which are determined by the shape of the coil, the number of turns of the insulated wire, the spacing between the turns, the magnetic constant, and the length of the coil.

The initial average value of the induction voltage is approximately 12 V. This value gradually decreases. At approximately 6 ms, the spool begins to move, as shown in Fig. 9. This is

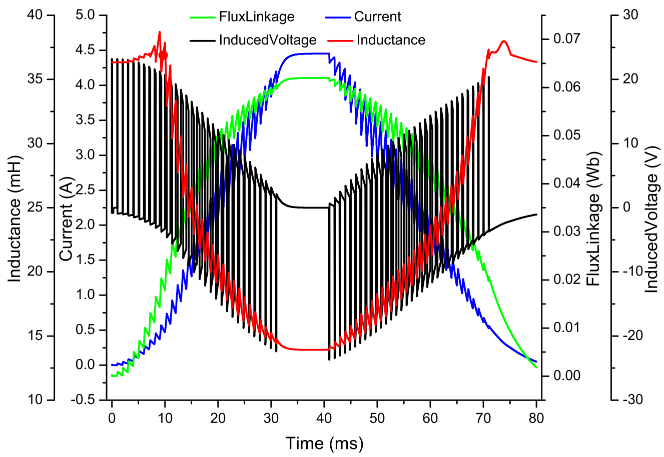


Fig. 7. Variation in amplitude of the coil current, the coil inductance, the flux linkage, and the induced voltage in the transient electromagnetic simulation.

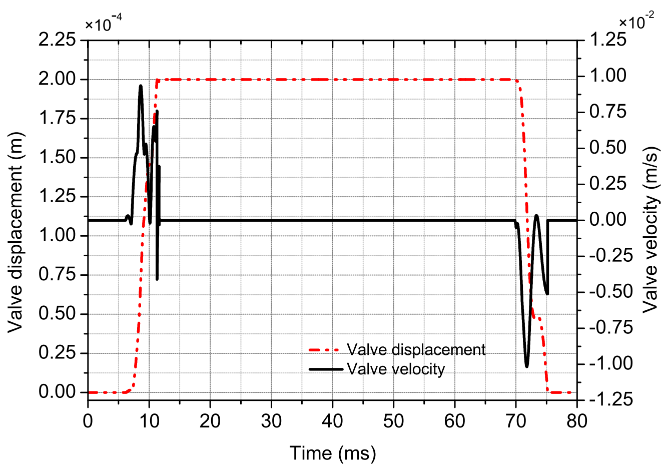


Fig. 8. Valve displacement and velocity.

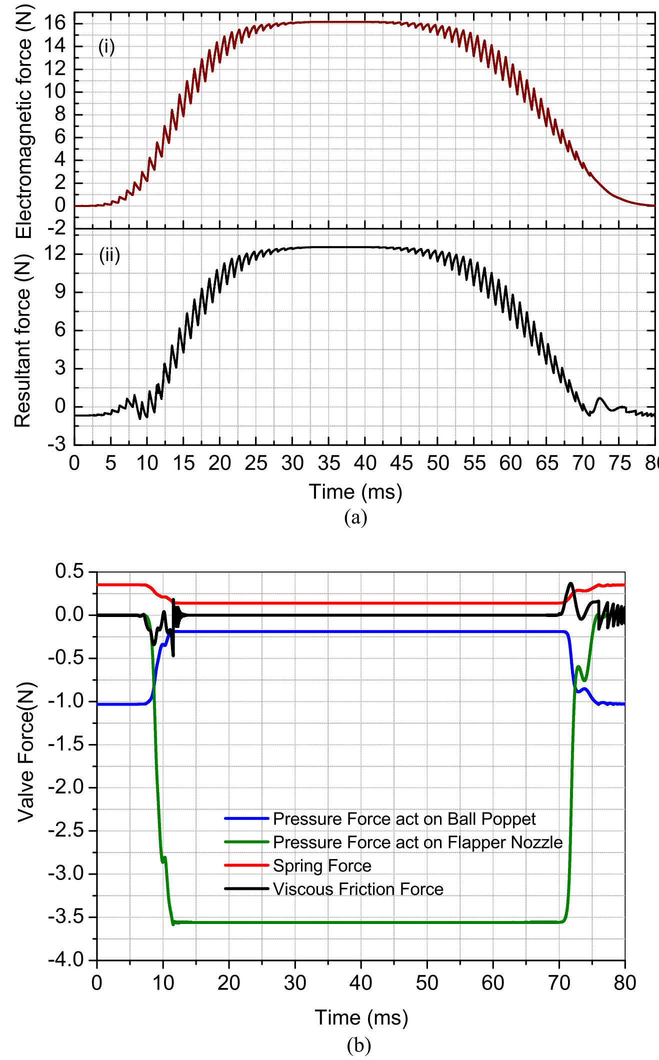


Fig. 9. Forces act on the valve in a transient simulation. (a) Electromagnetic force (i) and Resultant force (ii). (b) Pressure force, spring force and viscous force.

equivalent to a reverse electromotive force to the circuit and the current will have a downward trend, as determined in the literature[16], [37]. The spool will move to the maximum displacement (0.2 mm) at 12 ms, as shown in Fig. 8. The inductance dropped significantly at this stage from 37 to 30 mH. In the following phase, the spool stops moving and the duty cycle grows, the current increases, the magnetic flux increases, the inductance value declines, the induced voltage decreases to zero, and anti-to growth occurs. When the current reaches a maximum value of approximately 4.5 A, the maximum flux linkage is approximately 0.062 Wb, with an inductance of 14 mH, and the induced voltage is zero. Similarly, the gradual decrease of the duty cycle from 100% to zero can be analyzed with the understanding that each parameter performs opposite to the rise cycle; the current is reduced from 4.5 A to zero, the inductor value increases from 14 to 37 mH.

2) Forces and Motions: Fig. 8 represents the simulated time functions of valve body position and velocity. Fig. 7 shows that the valve begins to move at 6 ms when the current is approximately 0.2 A, which indicates that the valve dead zone is $0 \sim 0.2$ A. This maximum displacement (2×10^{-4} m) occurs at 12 ms when coil current reaches 0.9 A. The maximum velocity is less than 0.01 m/s at 8 ms not including the end-position. In the process of opening the PSV, there is a rise-fall-rise progression of the speed that is determined by the forces on the valve, as shown in Fig. 9(b). At 70 ms, the valve begins to close at a current of approximately 0.7 A. The current when the valve is completely closed is approximately 0.2 A. The entire process lasted 5 ms. The speed of the closing process is the opposite of that of the opening process with identical amplitude.

As mentioned earlier, dynamic forces acting on the spool promote spool movement. Fig. 9 shows the simulated time functions of the forces that act on the valve. Fig. 9(a) shows the electromagnetic force and sum of the forces act on the spool, and Fig. 9(b) shows the valve force, including the pressure force, acting on the ball poppet and flapper, the spring force, and the viscous force. Before the valve begins to move and as the current

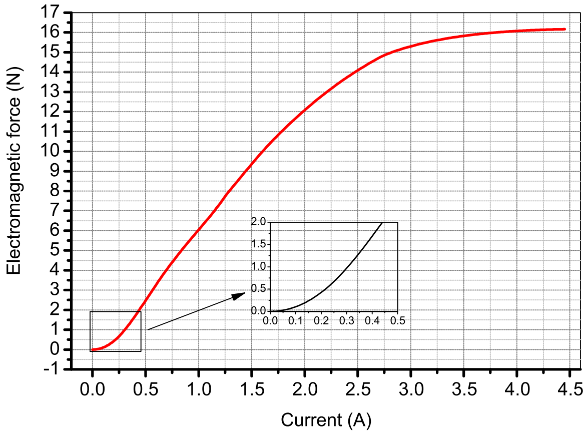
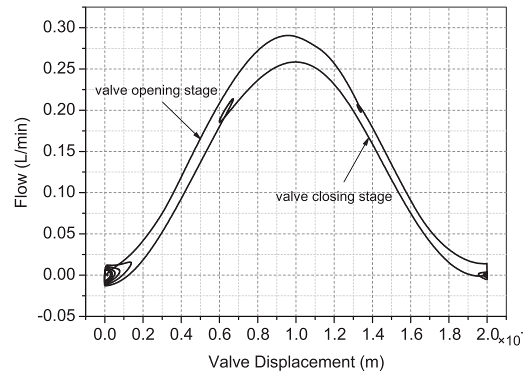
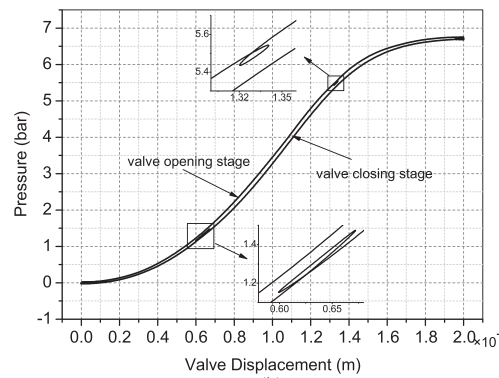


Fig. 10. Electromagnetic force versus the current in the transient simulation.



(a)



(b)

Fig. 11. Transient flow and pressure in the transient simulation. (a) Flow. (b) Pressure.

increases, the magnetic force increases, as shown in Fig. 9. The electromagnetic force cannot overcome the opposite resultant force (0.68 N) that contains the pressure force acting on the ball poppet (blue line, 1.03 N) and the spring force (red line, 0.35 N), as shown in Fig. 9(b). It also can be determined by Fig. 9(a), which shows that the resultant force acting on the valve is negative until the valve moves at approximately 6 ms. Because the valve is not open and there is no valve speed, both the pressure force acting on the flapper (green line) and the viscous force (black line) are zero.

Electromagnetic force continues to increase and the spool begins to move. The solenoid valve opens and the pressure force acting on the ball poppet decreases to 0.18 N from 1.03 N and the spring force from 0.12 N to 0.35 N. The pressure force acting on the ball poppet rises sharply, but the force growth rate is slower than the resultant force, including the pressure force, acting on the ball poppet. The spring force rate also decreases. This all occurs at approximately 8 ms, resulting in the decline shown in Fig. 9(b), as a result, the position-dependent force-dynamic characteristics fluctuate, resulting in an oscillation of valve velocity and displacement current (see Fig. 8). Here, the impact-function-based contact model is adopted to illustrate the force-dynamic characteristics when the flapper impact the valve body[38]. Because viscous force is negatively proportional to the velocity, the direction of viscous friction is converse to the speed. The maximum value (3.58 N) of the pressure force acting on the ball poppet occurs at 12 ms. The valve body performs a repetitive collision at its end-position because of viscous friction. The simulation has properly resolved the oscillations in velocity and in viscous friction. As shown in Fig. 9(b), the valve closing process is identical to the opening process, but the parameters change in the opposite direction.

In this system, as the current increases, the electromagnetic force increases significantly as shown in Fig. 10. There is a nonlinear relationship between the values of electromagnetic force and the current in the initial 0.2 A, as shown in the enlarged rectangle in Fig. 10. It presents a generally linear relationship between the current and the electromagnetic force in the range from 0.2 A to 1.5 A. The range encompasses the PSV working range (0.2 to 0.9 A). The electromagnetic force reaches a maximum of 16.2 N at the current $I = 4.5$ A. Figs. 9(a) and 10 verify this.

3) Fluid Dynamic Field: Assuming constant temperature, the output flow depends on the average spool position and on the supply pressure. Consequently, with constant supply pressure, the flow rate and pressure is related to the average spool position. Fig. 11 represents the valve dynamic flow rate of the discharge port and the pressure output during the transient simulation. The transition from close-open-closed position is considered. In the valve opening stage, the flow rate and pressure are gradually increased to 0.29 L/min and 3.5 bar, respectively, at a valve position of approximately 1.0×10^{-4} m. The flow rate is then gradually decreased and the pressure increases until it reaches a maximum valve of 6.7 bar at a valve position of 2×10^{-4} m. The flow rate drops to zero because the differential pressure reaches zero between the input and output of the discharge port. During the valve closing stage, the flow and pressure transform is opposite to those at the opening stage. The flow gradually increases whereas the output pressure declines, and the drain port peaks at 0.26 L/min with a displacement of 1.0×10^{-4} m because of the differential pressure at the drain port. Both the pressure and flow decreased to zero at the initial displacement.

Hysteresis can be observed in the pressure and flow data from Fig. 11(a) and (b). It is caused by the hysteresis material in the mover and the friction of movement. The maximum pressure and flow gaps were 0.2 bar and 0.03 L/min, occurring at a valve position of 1.0×10^{-4} m. Note that there are pressure

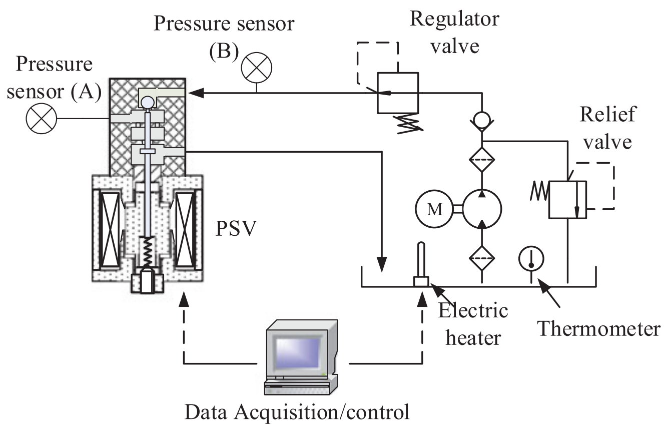


Fig. 12. Schematic diagram of the experimental setup.

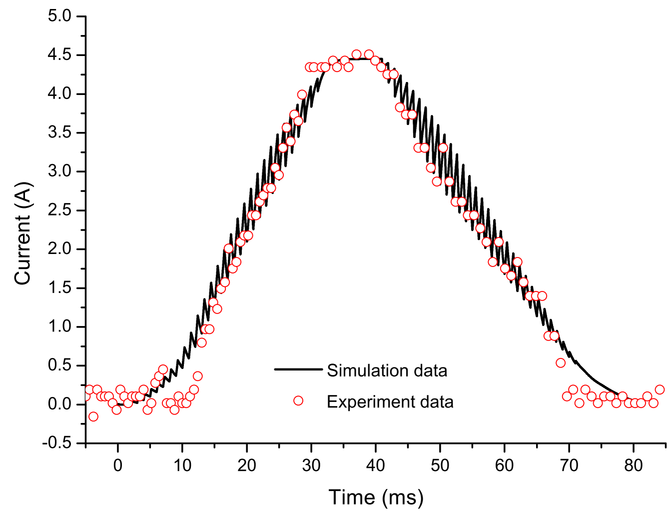


Fig. 13. Measured and simulated current in the coil.

and flow oscillations for the opened end-position and closed end-position limits. The closed end-position significant oscillation is caused by the spring, which is different from the opened end-position when the valve body penetrates the flexible contact surface. It is also remarkable that the valve displacement oscillation resulting from the position dependent force-dynamic characteristics fluctuates, leading to an oscillation of the pressure and the flow rate at the time of displacement oscillation. Fig. 11(b) enlarges the rectangle marked in Fig. 11(b) to clearly show the oscillation of the pressure.

V. Experimental Studies

A. Experimental Set-Up

To analyze the dynamic characteristics of the PSV and to verify of the above simulation model, a hydraulic test bench is set up. The schematic diagram is shown in Fig. 12. The PC with the ADC card samples stores the current, pressure data, and produces the signal as the control input to the modulator. The modulator which is an electronic circuit designed and constructed specially for these experiments, modulates the control signal. The output of the modulator is fed to the amplifier (ST VN540) and converted into a current signal.

A pressure compensated hydraulic supply with 10 L/min flow rate at 15 bar is used to power the system. Because there is sufficient compliance between the pump and the PSV, noticeable pressure fluctuations are not observed during the experiments. The pressure is measured by two pressure sensors (GEMS Pressure transducer) labeled A and B at the inward and outward ports of the valve, respectively. Automatic transmission fluid from Kunlun Lubricant is used for the experiment at the standard 20 °C evaluation temperature. The oil temperature is controlled using a 10 kW electrical heater and a 0.2 kW cooling pump with a fan. The controlling PWM period and duty cycle are identical to those in the simulation. The test is implemented for a 1 h interval while maintaining constant solenoid valve temperature. The power to the system is supplied by a constant pressure. The experimental process of the PSV is nonloading, and the output port is connected to an empty cavity. Thus, the mass and the viscous friction of the valve are the only loads driven by the system. The actuator has negligible internal leakage. Rigid connecting lines are used between the pump and the valve. The displacement is not measured due to limitations in the capabilities of the experimental facility.

B. Electromagnetic Verification

The electromagnetic force cannot be measured for this study because of the small dimensions. As a simple verification, the current is logged during the valve action process with the control strategy as the simulation. Fig. 13 shows the measured current (with an approximate 0.5% measurement error) of the actual solenoid together with the coupled simulation current. The sample time is 10 kHz. Due to the large quantity of the test data, only a few parts are selected to illustrate the results. It should be noted that the current curves of Fig. 13 are obtained by applying a 24-V input, which is the standard working voltage for a solenoid valve. The results indicate proper agreement between the two curves and verify the validity of the model.

C. Pressure Recording

To obtain the desired pressure versus duty cycle PSV behavior, Figs. 14 and 15 show the simulated and measured duty cycle functions of pressure for the PWM controlled PSV. It represents the measured and simulated dynamic output pressure of the solenoid valve with a 1000 Hz PWM input applied to the valve and the duty cycle varied from 0% to 100%. A constant supply pressure of 6.7 bar is maintained and the steady-state output flow of the valve is measured for any given duty cycle. According to the simulation and experiment results, at zero duty cycle, the solenoid does not move. The pressure begins to rise from zero at approximately 5% duty cycle. The pressure then gradually increases and reaches the maximum 6.7 bar at approximately 18% duty cycle, as shown in Fig. 15. When the solenoid valve begins to close at 14%, the output pressure begins to decline. The pressure decreases to zero at 3%, as shown in Fig. 15.

The simulation results are in good agreement with the experimental data from a mechanical viewpoint (pressure) and from

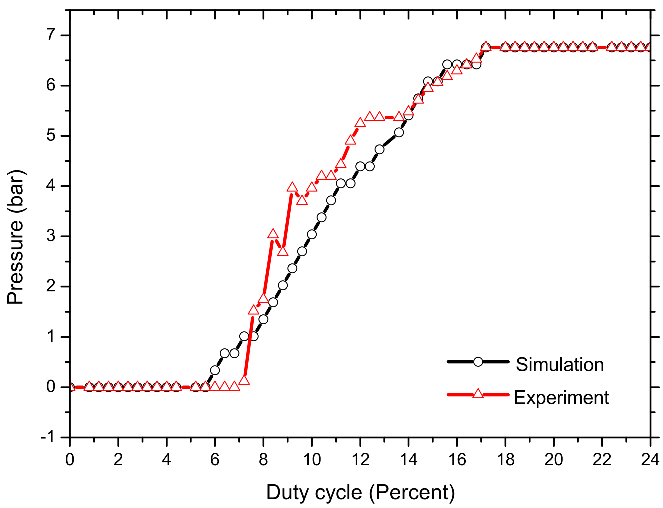


Fig. 14. Measured and simulated output pressure (up) of the valve.

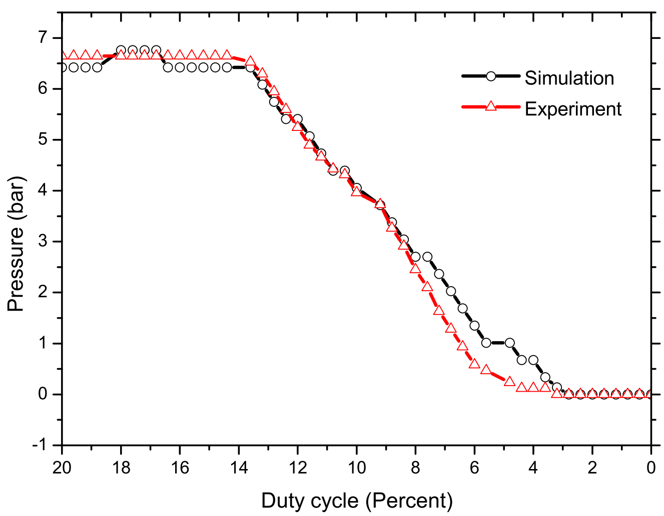


Fig. 15. Measured and simulated output pressure (down) of the valve.

an electromagnetic viewpoint (solenoid current), as shown in Fig. 13. It is also observed that there are deviations in the transient period of the two curves. This may be a result of neglecting the dynamic effects when modeling the connecting hoses, which affects the transient state of the system. Considering the resulting figures, it can be concluded that the PSV has an excellent ability to output a desirable pressure by controlling the flow from the discharge port. The transient response indicates a condition in which both the opening and closing times are within the appropriate range.

VI. Conclusion

An integrated coupling model has been presented for a PSV system in which the electrical, electromagnetic, fluid dynamic, and mechanical subsystems are strongly coupled. The analysis results demonstrate the dynamic performance under the PWM control and relationship between each parameter in the PSV system. The effectiveness of the model and analysis methodology has been verified by experiments.

The simulation results predicted that the PSV would initially open at a current of i= 0.2 A and require 6 ms to actuate with a duty cycle of 5%, which indicates that a dead zone exists at 0 – 0.2 A. It needs 12 ms and the i= 0.9 A with a duty cycle 18% when the PSV completely opening. The output is stayed linear with the controllable current within the range of 0.2A – 0.9 A. An instantaneous flow of 0.29 L/min resulted in the maximum output of 6.7 bar. The results indicate that the PSV own higher control precision and higher response time.

By reviewing the approach of the solution procedure, it was clear that the simulation environment made it possible to conveniently and effectively supplement the proposed PSV model while considering the fluid force and the electromagnetic force acting on the valve. This was performed using a reliable model of a fluid transmission that utilizes a PSV design. The methodology developed could also be used for PSV control algorithm optimization for PWM controlled PSVs.

Appendix A

TABLE I
Parametric Values Required for the Simulation

Physical Meaning	Notation	Unit	Valve
Input voltage	U	V	24
Resistance	R	Ω	5.2
Number of coil turns	N	turn	500
Electrical conductivity	σ_{coil}	S/m	6×10^7
Cross area of coil domain	A_{coil}	m^2	7.2×10^{-5}
Relative permeability	μ		1
Spool mass	m	Kg	0.062
Maximum spool stroke	$x_{\max}$	m	2×10^{-4}
Spring constant	k_s	N/m	1062
Initial compression	x_{s0}	m	3.3×10^{-4}
External valve diameter	dp	m	1.5×10^{-3}
Clearance on diameter	dc	m	5×10^{-6}
Length of viscous contact	l_c	m	1.8×10^{-2}
Supply pressure	P_s	bar	6.7
Oil density	ρ	Kg/m^3	850
Diameter of the flapper	dm	m	3.2×10^{-3}
Rod diameter(nozzle side)	dn	m	1.5×10^{-3}
Nozzle internal diameter	dr	m	2.8×10^{-3}
Ball diameter	da	m	1.8×10^{-3}
Seat diameter	ds	m	1.4×10^{-3}
Rod diameter (seat side)	dt	m	6×10^{-4}
Flow dynamic viscosity	u	Pa·s	0.03

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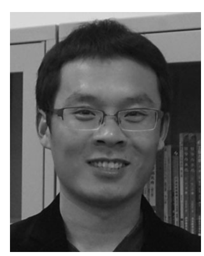

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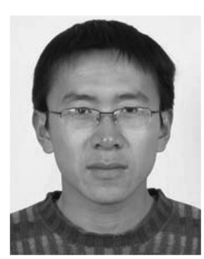

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